

Sleep, Body, Weight and Stress: A Look at the Sleep Health and Lifestyle Dataset

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Date: October 14, 2026

Introduction

Sleep duration is often associated with body weight and stress, but how strong is the relationship between them, and can it be demonstrated with a quantitative analysis? This project uses a single public survey dataset to ask two specific questions: is sleep duration related to BMI category and does self-rated stress levels predict how long people sleep? Both questions use the same dataset; therefore, a group-based factor (BMI category) and a numeric factor (stress) can be compared side by side.

Data sample

The dataset "Sleep Health and Lifestyle", a public survey distributed via Kaggle, includes 374 adults with 13 recorded variables, such as sleep duration, BMI category, sleep disorder diagnosis, blood pressure, sleep quality, etc. This project uses three columns: sleep duration (in hours), stress level (on a 1-10 self-rated scale) and BMI category. The raw file contains four labels for the BMI category, instead of three, because the "Normal" and "Normal Weight" categories both describe the same group. These two labels were merged into one category before analysis. After merging, the three groups are Normal (216 people), Overweight (148 people) and Obese (10 people) [1].

Methods

This analysis had three main steps: group, compare and correlate. For the first question, the 374 people were split into the three BMI categories and the median sleep duration was calculated within each group. For the second question, the relationship between stress level and sleep duration was measured with Spearman's rank correlation, which ranges from -1, for a perfectly opposite relationship, to +1, for a perfectly matched one. This does not assume a linear relationship between the two variables. All analysis was done in Python, using pandas for data handling, scipy for the correlation test and matplotlib for plotting.

Results and discussion

Median sleep duration is 7.5 hours for the Normal BMI group and 6.5 hours for the Overweight group, showing a difference of about one hour between the two (Figure 1). The Obese group has a median of 7.3 hours; however, since it only includes 10 people, this estimate is not as reliable as the other two and should be read with caution. Stress level and sleep duration show a strong negative correlation ($\rho = -0.81$). As self-rated stress rises from 3 to 8 on the 10-point scale, sleep duration drops from around 8.3 hours down to about 6 hours (Figure 2).

Conclusion

Both questions have a clear, quantitative answer in this dataset. Overweight adults tend to sleep less than those with a normal BMI, and stress level is strongly associated with sleep duration, as shown by the strong negative correlation. Future work could build on this by looking at BMI category, stress and physical activity together in one model.

References

[1] Sleep and Health Dataset. Available at: https://www.kaggle.com/datasets/uom190346a/sleep-health-and-lifestyle-dataset.

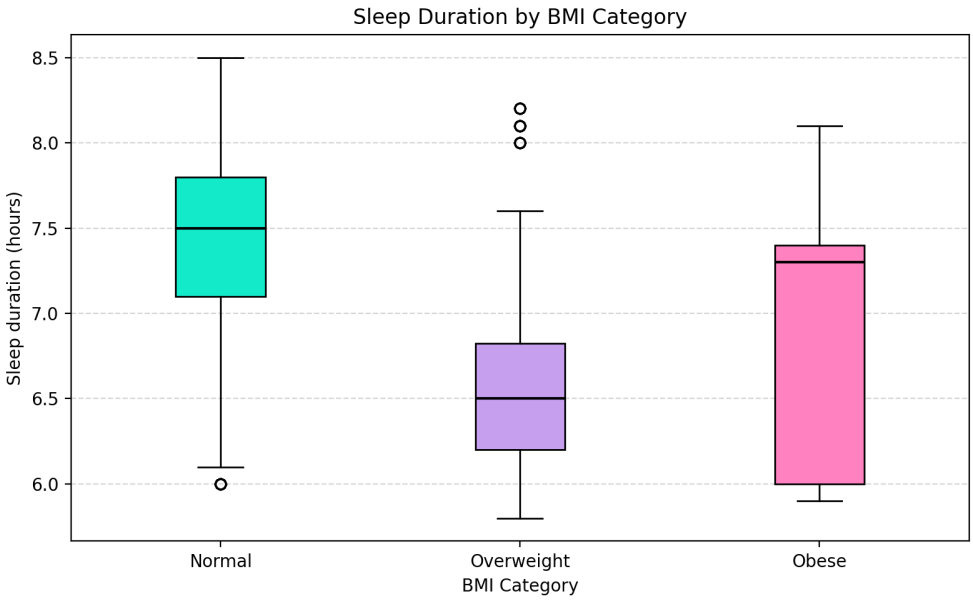


Figure 1: Sleep duration by BMI category

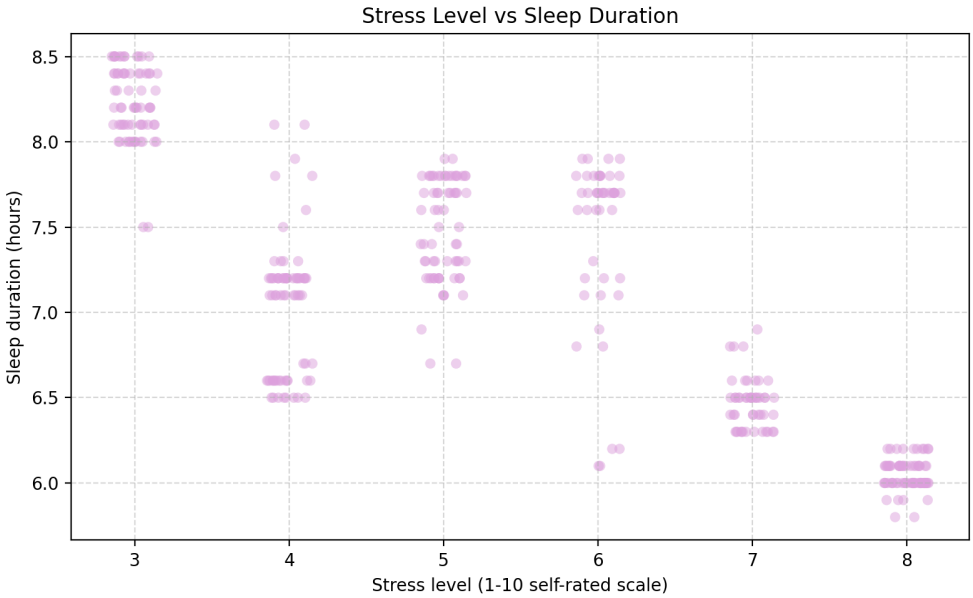


Figure 2: Stress level against sleep duration, each point one person